

Shivam Chaturvedi

PHD CANDIDATE

Indian Institute of Science, Bangalore

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Education

Indian Institute of Science

PHD CHEMICAL ENGINEERING

- Advisor: Prof. Ananth Govind Rajan

Bangalore

2022 - present

Malaviya National Institute of Technology

B.TECH CHEMICAL ENGINEERING

- Thesis advisor: Prof. Madhu Agarwal

Jaipur

2018 - 2022

Publications

PUBLISHED

Anand Mohan Verma*, **Shivam Chaturvedi***, Swastik Paul*, Ananth Govind Rajan and Others, *Data-driven massive reaction networks reveal mechanistic pathways underlying catalytic CO₂ hydrogenation*, **Nature Communications** (2026)

Shivam Chaturvedi, Abhijit Gogoi, and Ananth Govind Rajan, *Hydrocarbon Selectivity in CO and CO₂ Reduction on Al-Cu Single-Atom Alloy Electrocatalysts: Effects of Potential, pH, and Facet*, **ChemSusChem**, 19, e202501817 (2026).

Shivam Chaturvedi, Amar Deep Pathak, Nishant Sinha, and Ananth Govind Rajan, *Transient Microkinetic Modeling of Electrochemical Reactions: Capturing Unsteady Dynamics of CO Reduction and Oxygen Evolution*, **Advanced Theory and Simulations**, 9, e00799 (2025).

Raghavendra Rajagopalan*, **Shivam Chaturvedi***, Neeru Chaudhary*, Abhijit Gogoi, Tej S Choksi, Ananth Govind Rajan, *Advances in CO₂ reduction on bulk and two-dimensional electrocatalysts: From first principles to experimental outcomes*, **Current Opinion in Electrochemistry**, 51, 101668 (2025).

Bivas Bhaumik, **Shivam Chaturvedi**, Satyasan Changdar, Soumen De, *A unique physics-aided deep learning model for predicting viscosity of nanofluids*, **International Journal for Computational Methods in Engineering Science and Mechanics**, 24, 167–181 (2022).

IN PREPARATION

Shivam Chaturvedi, Nitish Govindarajan, Ananth Govind Rajan, *Copper's Missing Piece: A Machine-Learning and Graph-Theoretical Screening of Cu Single-Atom Alloys for CO₂ Reduction*

Anoop P. Pushkar*, **Shivam Chaturvedi***, Ananth Govind Rajan, *Hydronium and Water Proton-Transfer Pathways Together Govern pH-Dependent Selectivity in Electrochemical CO₂ Reduction*

* These authors contributed equally

Presentations

ORAL TALKS

Mar 2026. *Machine learning-guided design of copper-based single-atom alloy catalysts for CO₂ valorization* (Online) at the ACS Spring'26.

Jan 2026. *Advancing CO₂ Conversion via Data-Driven Pathways* at the IISc Chemical Engineering Symposium (ICS-26).

Jan 2026. *Unraveling Hidden Reaction Pathways in CO₂ Hydrogenation Using Machine Learning* (Online) at IIPE, Visakhapatnam.

Jan 2026. *Advancing CO₂ Conversion via Data-Driven Pathways* at the Faculty Development Programme (FDP), IISc, Bangalore.

Nov 2025. *Transient Microkinetic Modeling of Electrochemical Mechanisms: Towards Understanding the Dynamics of CO Reduction and Oxygen Evolution* at the AIChE Annual Meeting, Boston, Massachusetts.

Nov 2025. *Data-Driven Massive Reaction Networks Reveal New Pathways Underlying Catalytic CO₂ Hydrogenation* at the AIChE Annual Meeting, Boston, Massachusetts.

Oct 2023. *Microkinetic Modelling for Electrochemical Systems* at the Shell.ai Scientific Conference, Bangalore.

July 2023. *AI/ML for 2D Materials* at Namma Psi-k, JNCASR, Bangalore.

POSTER PRESENTATIONS

Dec 2025. *Transient Microkinetic Modeling of Electrochemical Mechanisms: Towards Understanding the Dynamics of CO Reduction and Oxygen Evolution* at ICCCU25, JNCASR, Bangalore.

Sep 2024. *Copper's Missing Piece: Screening Copper-Based Single-Atom Alloys for Electrochemical CO₂ Reduction Using Machine Learning* at the CARE Symposium, IIT Madras.

May 2024. *Copper's Missing Piece: Screening Copper-Based Single-Atom Alloys for Electrochemical CO₂ Reduction to Propylene* at the 2nd Divisional Research Symposium, IISc, Bangalore.

Oct 2023. *Development of a Microkinetic Framework for Electrochemical CO₂ Reduction* at the Shell.ai Scientific Conference, Bangalore.

Teaching Experience

Aug 2026 **NPTEL: Machine Learning for Core Engineering Disciplines**, Teaching Assistant under Prof. Ananth Govind Rajan

Aug 2025 **NPTEL: Machine Learning for Core Engineering Disciplines**, Teaching Assistant under Prof. Ananth Govind Rajan

Jan 2025 **CH253: Quantum-Mechanical Modelling of Nanomaterials**, Teaching Assistant under Prof. Ananth Govind Rajan

Aug 2024 **CH202: Numerical Methods**, Teaching Assistant under Prof. Narendra Dixit

Research Experience

Shell Technology Centre (STC)

Bangalore, KA

Co-ADVISORS: DR. AMARDEEP PATHAK, DR. NISHANT SINHA, AND PROF. ANANTH GOVIND RAJAN

Apr. 2023 – Oct. 2024

Unsteady-State Microkinetic Modeling of Electrochemical CO₂ Reduction

- Developed and tested a wrapper for MKMCXX, optimizing it for electrochemical CO₂ reduction simulations
- Built microkinetic models for reaction pathways and evaluated catalyst performance

Malaviya National Institute of Technology

Jaipur, RJ

ADVISOR: DR. MADHU AGARWAL

Aug. 2021 – Mar. 2022

Development of a Bio-Adsorbent for Arsenic Removal Using a Hydrothermal Process

- Developed orange peel-based bio-adsorbents for arsenic removal from potable water in rural India
- Conducted adsorption studies and characterized material performance

IASc-INSa-NASI Summer Research Fellowship Programme

Remote (Online)

ADVISOR: DR. VENKADACHALAM RAMESH (CENTRAL UNIVERSITY OF TAMIL NADU)

June 2020 – July 2020

Analyzing COVID-19 Spread in Chest X-rays Using CNNs

- Designed a CNN-based X-ray analysis model to predict COVID-19 with 96.9% precision and 91.7% recall
- Utilized ResNet with transfer learning for rapid and cost-effective disease detection

Institute of Technology & Management, Salt Lake

Remote (Online)

SUPERVISOR: DR. SATYASARAN CHANGDAR

Aug. 2020 – Feb. 2021

Predicting the Viscosity of Various Nanofluids Using Artificial Neural Networks (ANNs)

- Explored a physics-guided neural network framework to develop a generalized model capable of predicting nanofluid viscosity with high accuracy

Outreach

LEADERSHIP & SERVICE

2024-2025	Chemical Engineering Association , Vice President	<i>IISc</i>
2023-2025	Hockey Club , Club Convenor	<i>IISc</i>
2024-2025	Institute Innovation Council , Innovation Coordinator	<i>IISc</i>
2020-2021	NSS (National Service Scheme) , Core Team Member	<i>MNIT</i>
2018-2019	Energy Club , Executive Member	<i>MNIT</i>